ORACLE DB SETUP GUIDE

1. Download Required Software

- Download these:

Oracle 11g XE (Express Edition)

https://www.oicbasics.com/2020/01/download-oracle-database-11g-xe-express.html

Follow installation instructions and create a password, make sure to remember the password as this is your administrator password for Oracle database.

SQL Developer 24.3.1

https://www.oracle.com/asean/database/sqldeveloper/technologies/download/

Download **Windows 32-bit/64-bit,** and then extract the folder to desktop.

Java SE 17 (JDK 17)

https://www.oracle.com/java/technologies/javase/jdk17-archive-downloads.html

If you don't have **JDK 17 or above** installed, you need this as **SQL Developer requires** this. Select **Windows x64 Installer** to download. To confirm installation, open cmd and run
javac --version

Oracle Instant Client

https://www.oracle.com/asean/database/technologies/instant-client/winx64-64-downloads.html

Important for PHP and Oracle Database communication. To install, select **Version 11.2.0.4.0** then download the one that says **Instant Client Package – ODBC**. After installing extract it to **C:\oraclexe** (folder created when you install *Oracle*)

2. SQL Developer

- Open SQL Developer
- You will be prompted to enter the **JDK path**
- Find where **JDK** is installed in your system (usually at *C:\Program Files\Java\jdk-17*) and enter that.

- Create **New Database Connection** name it **SYSTEM** for username = system, and for password = (the one you set during **Oracle Installation**). Click 'Test', if success then 'Connect'.
- This is your **administrative connection** this is where you can do admin commands for your database. Oracle does not have multiple databases, it only has one database, and each connection is different views to the database.
- Next open the worksheet on **SYSTEM** (right click > Open SQL Worksheet)
- Create a new user through SQL queries, each user is its own schema, and each schema is its own workspace, essentially means each user is its own database.
- Enter and 'run script' this query one by one on the workspace:
    - `CREATE USER my_user IDENTIFIED BY my_password;`
    - `GRANT CONNECT, RESOURCE TO my_user;`
    - `ALTER USER my_user QUOTA UNLIMITED ON USERS;`
- Note: Replace my_user and my_password with your chosen username and password.
- Now create another connection for the new user, name it **ONE_QCU,** enter the username and password you just created. Test connection, then connect. You now have a new database workspace for standard user.
- Now to test your workspace, open workspace on **ONE_QCU,** and enter this query:
    - ```
      BEGIN
          DBMS_OUTPUT.PUT_LINE('Hello Oracle!');
      END;
      ```
- If you see an output then it's working.
- Useful queries:
    - `SELECT USER FROM dual;`
      // To see the current user selected on a connection
    - `SELECT username FROM dba_users ORDER BY username;`
      // To see all users on the database

3. Oracle 11g XE

- This is a checklist of configurations you need to setup in order for oracle to work.
- Go to **services.msc** (Win + R type it and run), among the list make sure that **OracleServiceXE** and **OracleXETNSListener** have their status to 'started', if not start it.
- Go to Win + R again and enter **sysdm.cpl** go to **Advanced > Environment Variables** then under **System Variables** find **PATH,** select then click edit. DO NOT delete Variable Value, instead go to the end of the text and type semicolon (;) and the path to your installed **Instant Client** which is usually at *C:\oraclexe\instantclient_11_2* Click 'OK' to all to save.
      ```

- Now to make sure that **ODBC** is enabled. Open **XAMPP** click **Config** and select **PHP (php.ini)** Inside this file make sure of the following:
    - `extension=oci8_11g`
      // Not used but add it anyway, search for OCI8 and add it with the other extensions.
    - `extension=odbc`
      // In order to enable, make sure this is not commented out, remove the semicolon (;)
    - `extension=pdo_odbc`
      // Enable it, remove the semicolon
- Then save.
- Restart your pc or just the **apache**.

4. Testing

- To test this, in this project folder there is a file name **test_db.php** run this in the browser. If it says **Database Connection Successful!** It means PHP and Oracle Database are able to communicate.

5. Create tables

- Open worksheet on **ONE_QCU.**
- Find the "_QUERIES" folder in this project.
- Open each, copy the queries, enter and run that query on the workspace.
- To view the tables go to **ONE_QCU,** select table and see if the tables are created (refresh if not).
- To test this, register an account to the **signup.php** page. Fill in the details and verify the email. If success, go to SQL Developer, select the 'users' table, around the top under tabs find 'Data' section and see if the account was added (refresh if not).

Note: You don't have to move the project out of htdocs, we are still using **xampp** but only the **apache** and not MySQL or phpmyadmin.

6. Additional Configurations and Fixes

- If you get this error:
    - `Database connection failed: SQLSTATE[IM002] SQLConnect: 0 [Microsoft][ODBC Driver Manager] Data source name not found and no default driver specified`

- Do this:
    - Win + R then type and run **C:\Windows\System32\odbcad32.exe**
- Switch to **System DSN** tab and check if the **OracleXE driver** is listed.
- If not, select **Add** > choose **Oracle in XE** > click **Finish**
- On **Data Source Name** type: **OracleXE**
- On **TNS** input field type: **XE**
- **Description** and **User ID** is not required
- If it asks for **Host** instead of **TNS:**
    - **Host:** localhost
    - **Port:** 1521
    - **Service Name:** XE
- Test Connection > input your created user credentials in oracle > 'Connection Successful'
- Restart **apache** then test on **test_db.php** again

Note: If **Oracle in XE** does not appear on the driver list it means: **Instant Client ODBC** is not installed, or **PATH** is not configured, or you have the wrong architecture. XAMPP and other apps must have the same architecture 64bit/32bit, if one is different, it will not work.